

# Gender and Task Alignment Influences the Credibility of Feminine but Not Masculine AI Conversational Assistants

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## Abstract

The present study investigated how the perceived gender of AI conversational assistants and the tasks they perform combine to influence perceptions of AI competence, trustworthiness, and goodwill (i.e., credibility). Participants evaluated AI assistants that appeared female or male and varied in the task they performed (stereotypically feminine or masculine task). Findings did not support an overall preference for female AI assistants over male AI assistants. Rather, perceived AI gender interacted with gender stereotypes associated with the task performed to affect perceptions of AI credibility. Notably, AI perceived to be very feminine had the highest credibility when performing stereotypically feminine tasks and the lowest credibility when performing stereotypically masculine tasks. The credibility of AI perceived to be masculine was not contingent on the task performed.

**Keywords:** gender, conversational agent, chatbot, artificial intelligence, trust

## Introduction

Artificial intelligence (AI) is revolutionizing communication and scholars are increasingly focused on understanding human-AI interaction (Markowitz, 2024; Wang & Gong, 2025; Zerfass et al., 2020). Specifically, the literature has widely examined anthropomorphism attributed to AI as a key variable of interest (Chinmulgund et al., 2023; Fakhimi et al.,

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2023). The frequent examination of anthropomorphism is a consequence of its constant presence in AI systems, generating many questions about its potential effects (Salles et al., 2020). Anthropomorphism, defined as an “attribution of human characteristics or behavior to inanimate objects” (Inie et al., 2024, p. 2322) can manifest with AI in many ways. For instance, manifestations can be represented by an AI voice (Kuch et al., 2023) or various appearance attributes (Crowell et al., 2019), with researchers examining how different aspects of anthropomorphism influence human-AI interaction.

A notable subset of AI scholarship investigates how gender is reflected in AI design and influences human-AI interaction (see Fortunati & Edwards, 2022). Scholars have documented the relationship between gender stereotypes and AI labor (Costa, 2018; Costa & Ribas, 2019), pointing out a massive feminization of AI systems (Antonopoulou, 2023; Costa & Ribas, 2019; Sutko, 2020). Such studies highlight the preference for feminine AI caused by social norms, gender roles, and stereotypes that frequently create an association between certain work tasks and gender (Ahn et al., 2022; Costa & Ribas, 2019). In other words, the feminization of AI is a consequence of the type of tasks that are commonly assigned to AI agents. For instance, AI is commonly associated with subordinate labor, and as part of a stereotyping process, individuals tend to create gender role expectations. Thus, a preference for systems perceived as feminine may emerge due to a linkage of common AI tasks with women’s labor (Costa & Ribas, 2019).

However, while the preference for feminine AI is documented in the literature, it remains unclear how or if this preference shifts when AI systems that appear feminine are placed in traditionally masculine contexts. Foundational CASA scholarship indeed suggests that people react more favorably to gendered computers that perform tasks that align (rather than misalign) with gender stereotypes (Nass & Brave, 2005). To more comprehensively understand how AI gender cues influence human-AI interaction, research is needed that considers how traditional gender roles related to work interact with AI gender cues. Thus, the present study aims to understand if preferences related to the perceived gender of AI (feminine vs. masculine) are moderated by the type of labor (stereotypically feminine or masculine) produced by AI agents. Consistent with some studies, feminine AI may be preferred despite the task performed due to users seeing the AI as warmer and more human (Borau et al., 2021). However, a preference for feminine AI might decline the more AI agents execute traditionally masculine labor (Ahn et al., 2022). Examining how gender stereotypes and task roles interact to influence human-AI interaction is essential to study in order to better understand (a) the ubiquity and magnitude of gender stereotype effects and (b) how users will experience AI systems.

## **Computers as Social Actors**

As outlined by Reeves and Nass (1996), the CASA framework posits that individuals’ interactions with computers are, at their core, social in nature. Thus, people perceive computers as independent agents, distinct from their programmers (Sundar & Nass, 2000), and interact with them as if they were social beings. Building on this premise, researchers have examined specific cues that shape computer social-driven perceptions. For instance, Nass and Brave (2005) review how variations in computer agents’ voices affect individuals’ responses. They discuss how people engage with computers through gendered voice-based

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cues. Specifically, some evidence exists that women prefer communicating with feminine-voiced computers, whereas men prefer communicating with masculine-voiced ones—a gender matching effect.

Other foundational studies suggest a preference for gender and task stereotype congruence. Nass et al. (1997) note how computer voices are subject to gender task stereotypes. Masculine-voiced computers are perceived as more competent in technical environments, whereas feminine-voiced computers are rated more empathic in relational contexts. In the realm of e-commerce, findings indicate that computer voices (masculine vs. feminine) exert different persuasive power when they promote products that align with gender stereotypes. For instance, there is evidence for a preference for masculine voices recommending stereotypically masculine products (Lee et al., 2000) and feminine voices recommending stereotypically feminine products (Nass & Brave, 2005). By laying the groundwork for understanding how gender is projected onto machines, these previous works pave the way for contemporary studies on gendered AI research.

The ongoing development of AI, however, has created systems with increasingly advanced human-like cues (Gambino et al., 2020). Such cues extend beyond feminine versus masculine voices, opening space for studies that explore additional characteristics such as avatars and names. These expanded cues suggest that AI-gendered perceptions may operate across a broader range of human-like characteristics than those previously documented (Nass & Brave, 2005; Nass et al., 1997). From this perspective, the present study aims to advance previous work on gendered computers by assessing the effects of different gendered AI pronouns, avatars, and names on judgments of artificial intelligence credibility. By bridging foundational CASA insights on congruency with contemporary AI evaluations, this study intends to provide a necessary update on how gendered perceptions operate within today's more socially expressive AI agents.

## Feminization of AI

Artificial intelligence and gender cues (feminine vs. masculine) are often developed within a feminization process (Costa & Ribas, 2019). In other words, AI features, such as voice tone, design characteristics, and names are constantly associated with femininity perceptions (Hester, 2017). Such perceptions of femininity are a consequence of intentionally designed systems that aim to be perceived as docile, servile, and replaceable (Sutko, 2020). Such designed systems reinforce gender stereotypes, leveraging femininity to make AI seem more approachable and compliant, and thus, more accepted (Sutko, 2020).

From a technological standpoint, there is no reason for AI systems to reflect a particular gender; the use of gender cues is a user experience strategy designed to create AI personas that convey familiarity and approachability (Costa & Ribas, 2019). Such choices, which have also opened debates regarding the collaboration of AI on negative gender stereotypes and social dynamics (Hilcenko & Taubman-Bassirian, 2023), are a consequence of cognitive processes which associate AI femininity with a more human-like interaction. In simpler terms, studies highlight that the use of gender stereotypes in AI systems strongly influences how human-like AI is perceived to be (Borau et al., 2021). Furthermore, beyond general assessments of being humanlike, feminine traits in AI are key predictors

of perceived warmth, empathy, and emotional intelligence (Hwang & Won, 2022), which positively strengthens human-AI relationships.

The familiarity and approachability of feminine AI may be observed in different contexts. For instance, Mirwan et al. (2023) through a descriptive analysis, indicated that feminine AI is considered more “approachable” and “pleasant” to interact with. Similarly, studies investigating the voice of AI assistants found that feminine voices are usually perceived as “warmer” and more acceptable (Mahmood & Huang, 2024). Lastly, Hwang and Won (2022) through a mixed-methods approach found that AI perceived as feminine is considered more socially intelligent and nonthreatening, which consequently, generates higher trust and engagement. These studies explore preferences for feminine AI systems across different contexts. Taken together, they appear to indicate that users will perceive feminine AI agents to be more credible than masculine AI agents as social intelligence, trust, and warmth respectively reflect components of credibility: competence, trustworthiness, and goodwill. Thus, we propose the following hypotheses:

**H1a:** Female AI assistants are rated as having more competence than male AI assistants due to greater perceived femininity.

**H1b:** Female AI assistants are rated as more trustworthy than male AI assistants due to greater perceived femininity.

**H1c:** Female AI assistants are rated as having more goodwill than male AI assistants due to greater perceived femininity.

## AI and Labor Type

As previously indicated, a common explanation for why users prefer feminine AI is that AI agents tend to perform tasks that are stereotypically associated with women (Schiller & McMahon, 2019). For instance, devices frequently used for daily task management such as *Alexa*, *Google Home*, and *Siri*, are consistently referred to as female agents (Hester, 2017; Schiller & McMahon, 2019). Such an association is established because women have been traditionally positioned in subordinate work and caregiving occupations (He, 2024; Hester, 2017). Thus, since the type of labor performed by AI usually aligns with tasks traditionally associated with women’s labor, AI systems are often designed with feminine characteristics. This approach aims to strengthen human-AI relationships by incorporating familiar gendered cues (Hester, 2017).

Thus, historical gender roles have led to the creation of a disproportionately large amount of AI perceived as feminine (He, 2024). However, although there is a feminine AI design majority, some studies aim their efforts at uncovering nuances related to the differences between AI perceived to be feminine or masculine. For instance, Moradbakhti et al. (2022) examined how users responded to personal banking AI that varied according

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1. We measure femininity and masculinity (i.e., gender expression perceptions) but created male and female AI assistants based on phenotypic cues. Thus, we predict and report on perceptions of gender but refer to stimuli and conditions as the female and male AI assistant conditions.

to gender and agency (high vs. low) and found that the male AI was more accepted for tasks that demand leadership. Furthermore, while using a high-agency female AI, participants identifying as men felt threatened during the interaction, but such an effect did not occur for women users. Ahn et al. (2022) explored how the gender of AI and the types of recommendations they produce influence user evaluations. They found that participants preferred recommendations for hedonic products (associated with emotion) to come from a female AI voice and recommendations for utilitarian products (associated with practical purposes) to come from a male AI voice.

Overall, the existing research on gender preferences for AI is mixed, with some studies suggesting a general preference for feminine AI regardless of the task performed and others suggesting that people prefer a match between the perceived gender of the AI and the gender stereotypes surrounding the role/task performed—similar to foundational CASA studies. Creating more complexity, some work indicates that users evaluate male AI performing masculine tasks as more competent, trustworthy, and useful than female AI performing feminine tasks (Crăiuț, 2023). In other words, the expectation of synchronicity between gender and labor stereotypes (Hester, 2017) was not found as a masculine alignment and was preferred over a feminine alignment. Furthermore, the article results found that male AI performing a feminine task did not suffer the same negative perception as the female AI performing a masculine task, which once again goes against previous results that suggest a global preference for female AI. H1a to H1c predict a main effect of gender on credibility perceptions. However, given the mixed results in the literature, the following alternative hypotheses are presented that take into account the stereotypes that accompany the tasks AI perform and past CASA results that suggest a preference for gender-task congruence.

**H2a:** There will be an AI gender by task gender cleaved interaction such that AI will be rated as having greater competence when performing tasks that align with gender stereotypes.

**H2b:** There will be an AI gender by task gender cleaved interaction such that AI will be rated as more trustworthy when performing tasks that align with gender stereotypes.

**H2c:** There will be an AI gender by task gender cleaved interaction such that AI will be rated as having more goodwill when performing tasks that align with gender stereotypes.

## Embracement of Stereotypical Gender Roles

One individual difference worth examining pertains to the views users have toward traditional gender roles and dynamics. Notably, people can vary in the degree to which they exhibit hostile and benevolent sexism. Glick and Fiske (1997) define hostile sexism as an openly negative attitude toward women, particularly in those circumstances in which women defy traditional gender norms. The authors suggest that such sexism is driven by a reinforcement of the patriarchy in which women represent a threat to men. Men

view women as manipulative, deceitful, and powerful, constantly aiming to occupy a man's place. Research has investigated individuals with high scores in hostile sexism in different contexts. For instance, hostile sexism thrives in contexts in which traditional gender roles are challenged. Gaunt (2013) found that individuals high in hostile sexism perceive female breadwinners as a direct threat to male authority.

On the other hand, benevolent sexism is defined by Glick and Fiske (1997) as a subjective attitude toward women that reinforces gender roles. Different from hostile sexism, benevolent sexism intends to add women in a place of nurturing and pureness, and, above all, an individual in need of male guidance. Such subjective sexism tends to reward women who understand their supportive societal roles, discouraging independence. It also tends to reinforce the aversion of women in leading roles (Hideg & Ferris, 2016).

Overall, there is reason to suspect that the results of our hypothesis tests might vary depending on participants' level of hostile and benevolent sexism. Given that the logic supporting both H1 and H2 revolve around users holding gendered stereotypes, the predictions are most likely to be supported by those who rate higher in hostile and benevolent sexism as they possess strong expectations about women and how they should behave in society. As such we propose the following research question:

**RQ1:** Are the results of H1 and H2 moderated by participant hostile or benevolent sexism?

To contextualize our investigation, we test our hypotheses and research questions based on Western gender stereotypes (Barrett, 2021; Miller & Hayward, 2007) with a sample of United States residents.

## Method

### Study Overview

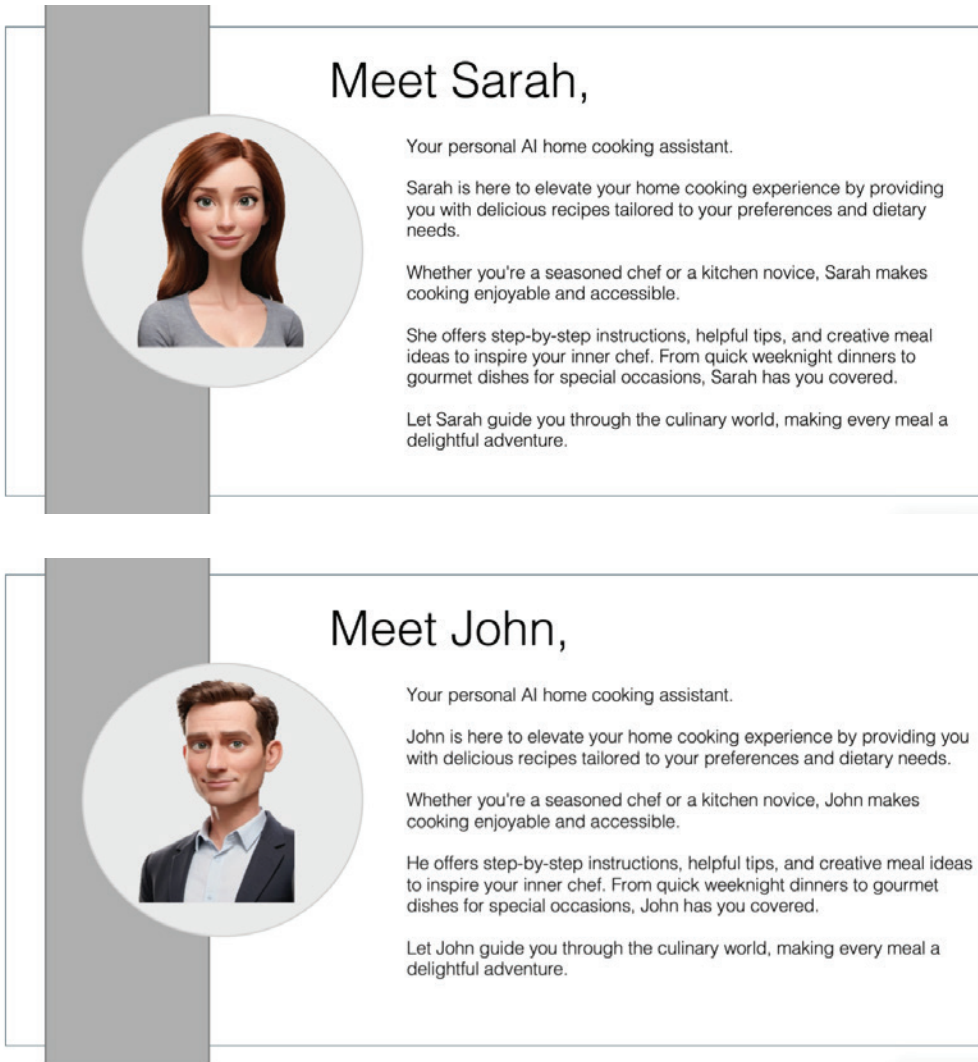
We designed a 2 (AI gender)  $\times$  2 (AI task) between-subjects experiment to test our hypotheses. Participants were randomly assigned to read about an AI assistant that varied in its gender depiction (female phenotypic cues vs. male phenotypic cues) and task performed (stereotypically feminine vs. masculine task; see Figures 1 and 2 for examples of each condition). After reading about the AI assistant, participants completed a questionnaire and received compensation. All study components were approved by the host institution's IRB.

### Participants

We obtained a sample of 403 participants from CloudResearch. Participants received \$1.50 for their efforts (mean completion time = 9 minutes and 22 seconds; median = 7 minutes and 54 seconds). Participants ranged from 18 to 68 years of age ( $M = 36.04$ ;  $SD = 10.90$ ) and identified as women (47.5%), men (50.1%), and nonbinary (2.2%)—one participant chose not to indicate their gender. The sample identified as European American, White, Anglo, or Caucasian (68.5%), African American, Black, African, or Caribbean (21.8%), Asian American, Asian, or Pacific Islander (6.9%), Hispanic American, Latino(a), or

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**FIGURE 1**  
**Examples of Stimuli Feminine Task Condition**

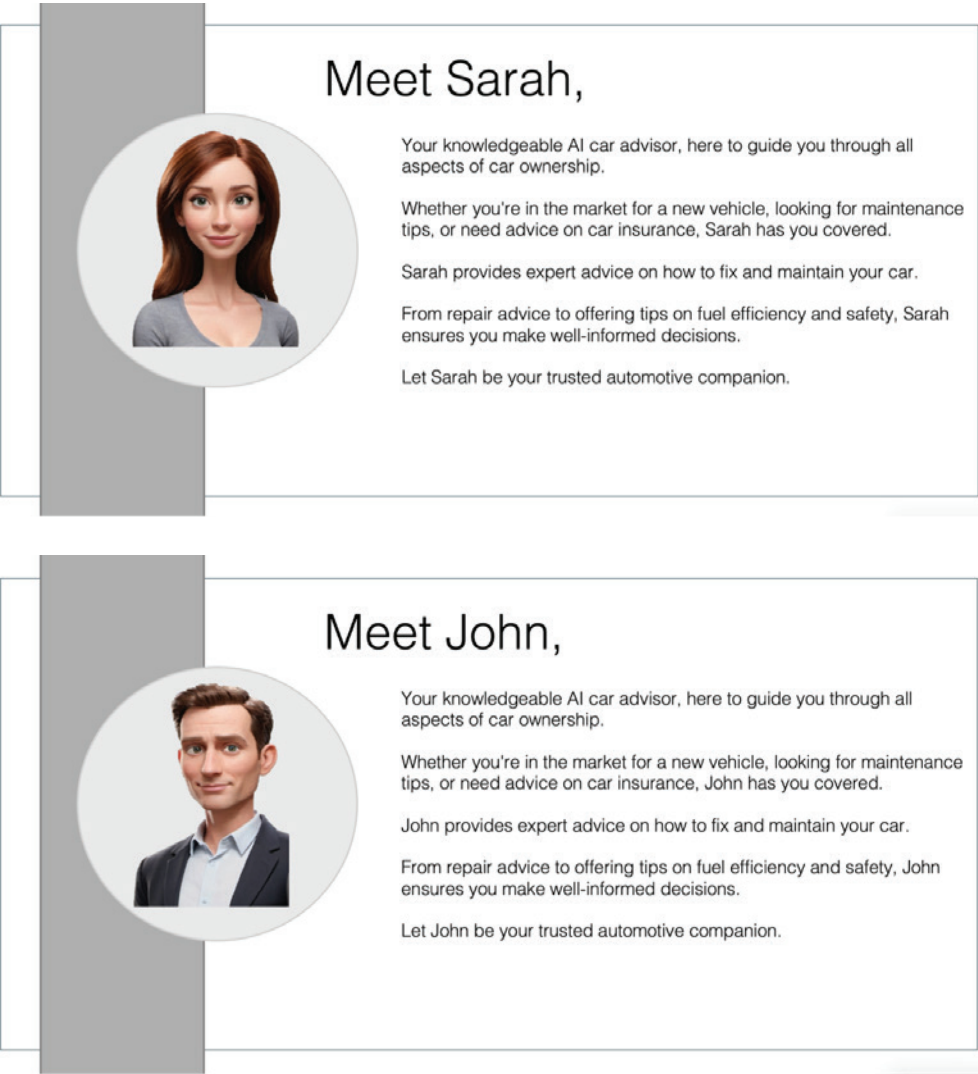


Chicano(a) (5.5%), Native American (1%), and of mixed background (.4%). Participants could identify with more than one race/ethnicity thus the overall percentage exceeds 100.

### ***Stimuli***

Across all conditions, participants viewed a description of a new AI assistant that varied in its appearance (female phenotypic cues vs. male phenotypic cues) and task performed (feminine or masculine gender role/task). Prior to viewing the stimulus and answering the questionnaire, participants read the following: “We would like your feedback on a

**FIGURE 2**  
**Examples of Stimuli Masculine Task Condition**



generative AI (artificial intelligence) chatbot assistant. You will be presented with a description and picture of an AI chatbot assistant. Then we will ask you questions about the AI chatbot and yourself.”

AI gender was varied by the appearance and name (Sarah vs. John) of the AI assistant. Specifically, we used phenotypic cues to create a prototypically female (Sarah) and prototypically male (John) avatar to induce perceptions that the AI visually represented a woman or man (see Lynch et al., 2024). Hereafter we refer to the stimulus as the female and male AI assistants. In an effort to promote generalizability, we selected multiple tasks that are associated with gender roles as indicated by Barrett (2021) and Miller and Hayward

(2007). Participants viewed a description of an AI assistant that performed *one* of the following tasks (depending on condition). The traditionally feminine tasks included AI assistants that provided help related to emotional support, home cooking, or nursing/health reminders. For the traditionally masculine tasks, an AI assistant provided help related to stock investing, automotive topics, or video games. Please see Figures 1 and 2 for examples.

## Measures

We chose to measure perceived AI gender and perceived task gender stereotype in a continuous fashion consistent with approaches by Wen et al. (2020) and Charlesworth and Banaji (2022). This allowed for greater nuance in assessing the degree to which AI gender perceptions and task gender stereotypes aligned or misaligned.

**Perceived AI gender.** The perceived gender of the AI was measured with a single-item on a 7-point scale. Participants were asked to rate the gender of the AI assistant on a scale that ranged from extremely masculine (1) to extremely feminine (7).

**Perceived task gender stereotype.** The perceived gendered nature of the task was measured with a single item on a 7-point scale. Participants were asked to indicate the extent to which they thought the task the AI would perform was a stereotypically masculine or feminine task on a scale from very masculine (1) to very feminine (7).

**Source credibility.** Source credibility was measured using J. C. McCroskey and Teven's (1999) credibility scale that assesses competence, trustworthiness, and goodwill. Six semantic differential items are used to assess each component of credibility on 7-point scales. The scales measuring competence (e.g., inexpert/expert;  $\omega = .94$ ), goodwill (e.g., insensitive/sensitive;  $\omega = .92$ ), and trustworthiness (e.g., dishonest/honest;  $\omega = .94$ ) were reliable. For decades this scale has been used to assess how perceptions of source credibility affect an array of important interpersonal outcomes (see Finn et al., 2009; L. L. McCroskey et al., 2014) and has recently been used to assess credibility in human-machine interactions (e.g., Abendschein et al., 2021).

**Ambivalent sexism inventory (ASI).** The ASI was used to measure hostile and benevolent sexism (Glick & Fiske, 1997). Each measure has 11 items assessed on 5-point scales ranging from strongly disagree to strongly agree. An example item for the hostile sexism scale is, "Women exaggerate problems they have at work" ( $\omega = .90$ ). An example item for the benevolent sexism scale is, "A good woman should be set on a pedestal by her man" ( $\omega = .84$ ).

## Results

Prior to examining the hypotheses and research question, we examined the degree to which our inductions operated as intended. The perceived gender of the male ( $M = 2.75$ ,  $SD = 1.16$ ) and female ( $M = 6.12$ ,  $SD = .75$ ) AI assistants varied as expected by condition,  $F(401) = 1205.96$ ,  $p < .001$ ,  $\eta^2 = .75$ . Likewise, the perceived gendered nature of the task (hereafter referred to as task gender) varied as anticipated with participants differentiating the masculine tasks ( $M = 3.31$ ,  $SD = 1.30$ ) from the feminine tasks ( $M = 4.57$ ,  $SD = 1.34$ ),  $F(401) = 92.19$ ,  $p < .001$ ,  $\eta^2 = .19$ . One-sample  $t$ -tests indicated that each masculine task—stock assistant ( $M = 3.55$ ,  $p = .005$ , Cohen's  $d = .35$ ), automotive assistant ( $M = 2.85$ ,  $p < .001$ , Cohen's  $d = .89$ ), and video game assistant ( $M = 3.53$ ,  $p = .003$ , Cohen's  $d = .39$ )—was

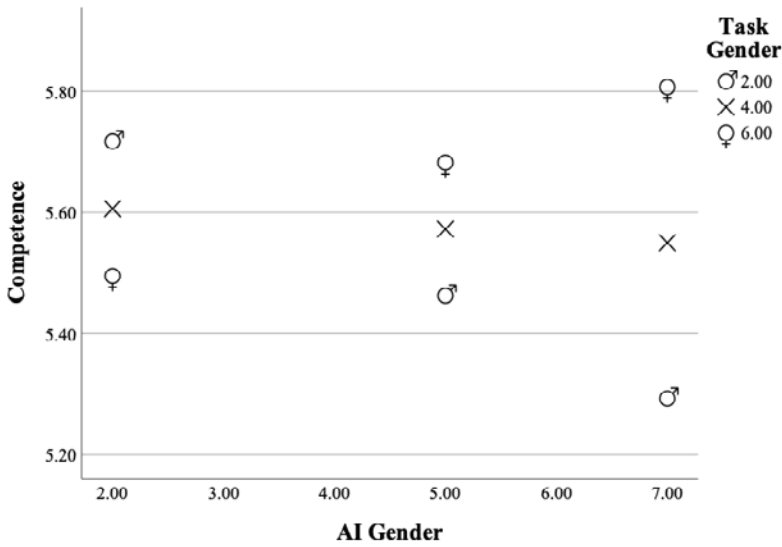
rated below the scale midpoint of 4. Likewise, each feminine task—emotional support assistant ( $M = 4.67$ ,  $p < .001$ , Cohen's  $d = .51$ ), home cooking assistant ( $M = 4.48$ ,  $p = .008$ , Cohen's  $d = .33$ ), and nursing assistant ( $M = 4.57$ ,  $p < .001$ , Cohen's  $d = .46$ )—was rated above the scale midpoint of 4. As such, all of the following analyses collapse the masculine tasks and the feminine tasks into a single masculine (coded 1) or feminine (coded 2) task gender condition.

H1a states that female AI assistants are rated as having more competence than male AI assistants due to greater perceived femininity. To test this hypothesis, we conducted a mediation analysis via Model 4 in PROCESS (Hayes, 2022). For all hypothesis tests, effect estimates are reported with percentile bootstrap confidence intervals based on 10,000 resamples. Using mediation (H1) and moderated mediation models (H2) allows us to directly examine the intervening social inferences that theory predicts will shape the outcomes of interest.

For H1a, the effect of the AI gender condition (X) on competence (Y) through perceptions of AI gender was estimated. We did not detect an indirect effect  $b = -.05$  ( $SE = .19$ ), 95% CI =  $[-.43, .31]$ . Indirect effects were also not detected for the outcomes of trustworthiness (H1b),  $b = -.12$  ( $SE = .22$ ), 95% CI =  $[-.56, .29]$ , and goodwill (H1c),  $b = -.02$  ( $SE = .25$ ), 95% CI =  $[-.54, .44]$ . Thus, H1a, H1b, and H1c were not supported. Increased perceptions of femininity did not enhance the perceived competence, trustworthiness, or goodwill of the AI assistant across all tasks. To address RQ1, we examined if hostile or benevolent sexism interacted with perceptions of AI gender to moderate the above relationships. No evidence of moderation was detected for any of the outcome variables (see Table 2).

H2a states that there will be an AI gender by task gender cleaved interaction such that AI will be rated as having greater competence when performing tasks that align with gender stereotypes. To test this hypothesis, we used Model 14 in PROCESS (Hayes, 2022) to detect moderated mediation. Specifically, we examined if perceived task gender moderated the effect of perceived AI gender on competence (H2a), trustworthiness (H2b), and goodwill (H2c). For the outcome of competence, significant moderated mediation was detected,  $b = .12$  ( $SE = .06$ ), 95% CI =  $[.02, .24]$ . A pattern consistent with cleaved moderation (see Holbert & Park, 2020) was detected (see Figure 3 and Table 1). Significant moderated mediation was also detected for the outcomes of trustworthiness,  $b = .24$  ( $SE = .07$ ), 95% CI =  $[.11, .39]$ , and goodwill,  $b = .27$  ( $SE = .09$ ), 95% CI =  $[.10, .44]$ . However, the nature of the significant moderation was more consistent with contingent rather than cleaved moderation (see Figure 3 and Table 1). For tasks perceived to be masculine, the perceived trustworthiness and goodwill of the AI assistant was reduced the more the AI was perceived to be feminine. Although the inverse trended in the predicted direction (i.e., more trust and goodwill for AI performing feminine tasks, the more the AI is perceived to be feminine) the conditional indirect effect was not significant. To address RQ1, we examined if any of the significant effects reported above varied depending on self-reported levels of hostile and benevolent sexism. No significant effects emerged (see Table 2). As a final robustness check, we examined if self-reported participant gender affected any of the hypothesis tests. Participants who identified as women and men did not differ in how they evaluated the gender of the AI ( $p = .49$ ) or task gender ( $p = .30$ ); participant

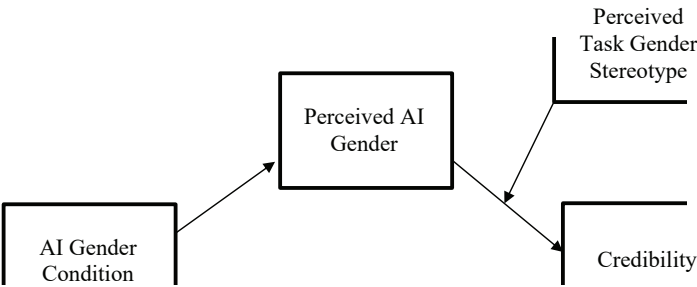
**FIGURE 3**  
**Conditional Indirect Effects for Outcomes of Competence**



**TABLE 1**  
**Indirect Effect of AI Gender on Outcome Variables at Different Levels of Task Gender Role**

| Path Estimates<br><i>b</i> (SE)            | Outcome Variables (Y)     |                             |                             |
|--------------------------------------------|---------------------------|-----------------------------|-----------------------------|
|                                            | Competence                | Trustworthiness             | Goodwill                    |
| Constant                                   | *6.05 (.35)               | *6.01 (.40)                 | *5.53 (.45)                 |
| AI Gender Condition (X)                    | .06 (.21)                 | .05 (.24)                   | -.05 (.28)                  |
| Perceived AI Gender (M)                    | -.16 (.09)                | *-.33 (.10)                 | *-.35 (.11)                 |
| Perceived Task Gender (W)                  | -.13 (.09)                | -.20 (.10)                  | -.14 (.12)                  |
| Perceived Gender x Task                    | *.04 (.02)                | *.07 (3.52)                 | *.08 (.40)                  |
| F                                          | 1.32                      | 5.20                        | 7.36                        |
| R <sup>2</sup>                             | .01                       | .05                         | .07                         |
| Conditional Indirect Effects (SE) [95% CI] |                           |                             |                             |
| At 16th percentile of task gender          | -.29 (.21)<br>[-.72, .12] | -.64 (.24)<br>[-1.13, -.18] | -.66 (.30)<br>[-1.29, -.11] |
| At 50th percentile of task gender          | -.04 (.21)<br>[-.45, .37] | -.16 (.23)<br>[-.61, .28]   | -.13 (.24)<br>[-.62, .33]   |
| At 84th percentile of task gender          | .21 (.26)<br>[-.29, .74]  | .32 (.30)<br>[-.23, .95]    | .40 (.30)<br>[-.18, .98]    |
| Index of Moderated Mediation (SE) [95% CI] |                           |                             |                             |
|                                            | .12 (.06)<br>[.02, .24]   | .24 (.07)<br>[.11, .39]     | .27 (.09)<br>[.10, .44]     |

**FIGURE 4**  
**Visualization of Moderated Mediation Analyses**



Note. Model 14 in PROCESS (Hayes, 2022) used to estimate the effect of perceived AI gender on perceptions of credibility (competence, trustworthiness, and goodwill) at different levels of task gender stereotypes (i.e., tasks perceived to be masculine, neutral, and feminine). See illustration above in Figure 3.

**TABLE 2**  
**Indices of Moderated Mediation and Moderated Moderated Mediation for RQ1**

|                                            | Outcome Variables (Y)    |                           |                           |
|--------------------------------------------|--------------------------|---------------------------|---------------------------|
|                                            | Competence               | Trustworthiness           | Goodwill                  |
| Index of Moderated Mediation (SE) [95% CI] |                          |                           |                           |
| Hostile Sexism H1                          | .06 (.10)<br>[−.14, .27] | .11 (.13)<br>[−.15, .35]  | .21 (.15)<br>[−.10, .49]  |
| Benevolent Sexism H1                       | .12 (.15)<br>[−.17, .41] | .09 (.17)<br>[−.24, .41]  | .26 (.19)<br>[−.13, .64]  |
| Hostile Sexism H2                          | .05 (.10)<br>[−.14, .26] | −.01 (.13)<br>[−.24, .26] | −.04 (.15)<br>[−.32, .28] |
| Benevolent Sexism H2                       | .09 (.14)<br>[−.17, .37] | −.02 (.17)<br>[−.35, .32] | .003 (.19)<br>[−.38, .38] |

gender was also not a significant covariate in any of the mediation models used to test H1 and H2 ( $p$ 's = .35 to .93).

Discussion

The present study investigated how AI-perceived gender and task stereotypes influence perceptions of AI credibility, specifically in terms of competence, trustworthiness, and goodwill. Participants inferred the gender of an AI assistant from an image and name and recognized tasks as being stereotypically more or less masculine or feminine. This confirms that people will ascribe gendered qualities to nonhuman entities and tasks they perform in a way that is consistent with their social interactions with others. Our first

set of hypotheses examined whether participants would have an overall preference for more feminine AI assistants regardless of the task the AI would perform. Our results did not support an overall preference for more feminine AI. Rather, and consistent with our second set of predictions, the perceived gender of the AI interacted with task gender stereotypes to influence the perceived credibility of the AI assistant. Notably, the nature of the interaction revealed a gender asymmetry. As detailed below, task incongruence diminished credibility for the feminine AI, whereas evaluations of masculine AI were largely unaffected by the type of gendered task performed.

Formally, the nature of the interaction effect varied across the three outcome measures of competence, trust, and goodwill. For competence, we detected a pattern consistent with cleaved moderation (see Holbert & Park, 2020). Competence increased the more the perceived gender of the AI assistant aligned with the gender stereotype of the task and decreased the more the perceived gender of the AI assistant diverged from the gender stereotype of the task. Again, the effect of congruence/incongruence was largely driven by the feminine AI being rewarded for congruence and penalized for incongruence; competency perceptions did not vary for masculine AI across task types.

For the outcomes of trust and goodwill, the nature of the moderation was more consistent with contingent moderation. The clearest pattern of results across all three outcome measures reinforced that congruence/incongruence was consequential for AI perceived to be very feminine; when tasks were also feminine, the highest levels of competence, trust, and goodwill were observed. Conversely, when the tasks were thought to be very masculine, the lowest levels of competence, trust, and goodwill were observed. In contrast, AI perceived to be very masculine were not viewed as more or less credible depending on whether they performed a masculine or feminine task. The generalizability of our results is enhanced by the fact that participants were randomly assigned to view one of three different tasks that are traditionally viewed as masculine or feminine; this helps reduce concerns that specific tasks (e.g., nurse assistant vs. gaming assistant) influenced the pattern of results.

Notably, the interaction effects reported above were not moderated by individual differences in sexism. The null results of self-reported levels of hostile and benevolent sexism might reflect broadly shared gender schemas among participants. Gender stereotypes are widely shared across men and women and are chronically salient, such that people readily apply them even when they are not explicitly endorsed (Ellemers, 2018). In addition, gender stereotypes are prescriptive, reflecting perceived societal standards about how women and men should behave rather than purely personal beliefs (Prentice & Carranza, 2002). Because these expectations are understood as common cultural knowledge, even individuals low in hostile or benevolent sexism may rely on them when making judgments. Thus, gendered evaluations of AI may be driven by pervasively held schemas that are broadly internalized; this may have limited our ability to detect effects attributed to variability in sexism in our sample.

The overall pattern of results suggests that the perceived gender of AI and the tasks AI perform should be considered in conjunction as these perceptions combine to influence how AI are evaluated. The data also suggest that considering stereotypes surrounding the task AI performs is more important for AI perceived as feminine. Put differently, there was

much more variability in how the feminine AI was evaluated relative to the masculine AI. Measuring perceptions of AI and task gender stereotypes in a continuous fashion allowed us to assess various points of alignment/misalignment to uncover these effects. Although participants, on average, differentiated the feminine and masculine AI and tasks as anticipated, there was variability in how strongly participants viewed the AI and tasks as feminine or masculine; this nuance would have been lost if we did not operationalize gender perceptions and stereotypes in a continuous fashion (see Wen et al., 2020).

## Theoretical and Practical Implications

The current findings lend support to theories in human-machine communication (HMC), and human-agent interaction (HAI), such as CASA (Gambino et al., 2020; Nass et al., 1997). As discussed by Reeves and Nass (1996), CASA challenges the assumption that human and machine interaction is solely driven by rational cues. In doing so, the framework argues that individuals place social rules and expectations on digital technologies as if they were human beings. For instance, in this study, participants inferred the AI's gender from characteristics such as its name and avatar, and then applied social stereotypes to evaluate its credibility, as if the AI assistants were human beings. AI assistants were judged as more competent when their perceived gender *matched* the gender stereotype of the task (e.g., masculine AI for masculine-coded tasks), and less competent when they diverged. Building on foundational work that examines voice-based gender cues (Nass & Brave, 2005; Nass et al., 1997; Reeves & Nass, 1996) we document how visual cues (e.g., avatars) can trigger gendered stereotypes and expectations, affecting the perceived credibility of AI.

Thus, our findings contribute to post-CASA theoretical discussions. Drawing on Lombard and Xu's (2021) Media Are Social Actors (MASA) paradigm, our results may inform how primary and secondary social cues can shape responses to AI. Specifically, altering the AI's facial appearance—a primary cue—significantly influenced participants' perceptions of its credibility. These facial cues were accompanied by a secondary cue: the language used to describe the AI, which employed feminine or masculine pronouns. The MASA paradigm offers an explanation for the observed differences in credibility perceptions between task and gender (Lombard & Xu, 2021). Although we did not directly measure processing (e.g., mindful vs. mindless processing), the observed results are consistent with this distinction. When the AI gender presentation aligned with the stereotypical expectations of the task, participants likely relied on mindless processing, automatically applying familiar social schemas that enhanced perceptions of credibility. In contrast, when the AI's gender cues conflicted with task expectations, the incongruence may have triggered mindful processing, making participants more aware of the AI's artificiality, and prompting deliberate evaluation. This shift toward reflective processing perhaps reduced perceptions of credibility, consistent with MASA's proposition that unexpected or mismatched social cues elicit more conscious, evaluative responses. Future work is needed to further explore and empirically test these possibilities directly.

Another central finding was that the evaluation of feminine AI assistants showed significantly more variability than that of masculine AI assistants across tasks. AI assistants

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perceived to be very feminine were rated *highly* when performing feminine-coded tasks but received the *lowest ratings* for competence, trust, and goodwill when assigned masculine-coded tasks. In contrast, masculine AI assistants were evaluated more consistently across task types. This asymmetry shows a gender-based double standard. Feminine AI assistants were more susceptible to stereotype-consistent or violating judgments, which echoes real-world gender bias in human evaluations. Research on gender stereotypes indicates that women are more susceptible to gender roles, thus being more rigorously punished in cases of divergence compared to men (Davidson, 1981). Such results, consistent with CASA, show the magnitude to which individuals view AI assistants as social actors.

In terms of practical implications, the results corroborate the idea that the perceived gender of AI should not be treated as an arbitrary design choice. Instead, it should be treated as a critical design element that directly impacts AI success. Whether AI gender cues align or conflict with the gendered expectations of the task AI performs needs to be taken into consideration. Additionally, these findings highlight how even minimal design elements—such as an AI's name and avatar image—can significantly shape and reinforce gender stereotypes. Notably, the mere presence of a name and picture, despite the AI not resembling an actual person, was enough to influence participants' perceptions of gender and what tasks they would be good at. This underscores an important point: Even abstract, cartoonish, or nonhuman representations can ingrain and/or reinforce social biases.

Finally, the results illustrate important paradoxes about aligning AI gender with task stereotypes. On one hand, such alignment may enhance the perceived credibility of AI. However, such alignment may also reinforce harmful gender stereotypes in digital spaces and, by extension, impact future human social interactions. For instance, research suggests children might be especially prone to establishing relationships with conversational agents such as *Alexa*, with such interactions influencing subsequent familial roles and dynamics (Sciuto et al., 2018). Rheu et al. (2021) provide a systematic review of an array of factors that influence trust in conversational agents. Although a complete review of conversational agent trust and AI ethics is beyond the scope of the current study, Fossa and Sucameli (2022) provide a thorough discussion of this precise ethical quandary.

Fossa and Sucameli (2022) note that many programmers and developers are focused on making their social robots and AI conversational agents as approachable and user-friendly as possible. Accordingly, they might employ gender cues to achieve such goals without direct consideration of the social biases they may be furthering. Fossa and Sucameli assess (a) the degree to which it is ethically permissible to exploit gender biases to improve human-machine interactions, (b) whether gender cues can be used so long as they do not promote *harmful* gender stereotypes, and (c) whether designers should be required to proactively promote ethical attitudes by combatting harmful gender stereotypes through their design of technologies intended for human-machine interaction. Many contextual factors might shape the response to these ethical questions. For instance, the ethical assessment of a hospital that relies on an AI assistant to ensure patients follow critical medical advice might vary relative to a company using a gendered AI assistant to sell more car insurance. Likewise, the ethical assessment of aligning gender cues with task stereotypes might vary for initial interactions with populations who (a) distrust technology but (b) experience disparities that an AI agent might help overcome (if only the users trusted the agent). Such ethical considerations can be enriched by Butler's (2015) theory of gender performativity

that, among other things, examines how gender reproduction is always a negotiation with power.

## **Limitations and Future Directions**

There are limitations in our study which should be considered and addressed in future research. For instance, we focused only on AI perceived to reflect women or men. We made this decision in order to test predictions wherein past work found certain tasks to be clearly linked to masculinity or femininity. Future work should adopt a more expansive and nuanced approach to gender and stereotypes, especially as stereotypes emerge that align with other gender identities. The use of a sample of United States residents and a focus on Western gender stereotypes should be taken into account when considering the generalizability of our results. Interculturally, there most certainly is variability in (a) stereotypes associated with gender, (b) how closely adherence to cultural standards is scrutinized, and (c) the social and/or legal costs of misalignment. Future research should explore how cultures with strict gender role expectations react to anthropomorphized AI that violate cultural mores.

Furthermore, we did not have participants interact with actual AI assistants. As such our findings might best inform how people react to initial descriptions of AI based on gender stereotypes. The degree to which initial impressions might persist or quickly dissipate based on actual interaction is unclear. Future research should explore the conditions under which initial impressions greatly influence evaluations of AI assistants and a willingness to use them as well as the conditions that might make initial opinions fleeting.

Additionally, work building on our results should further examine why there was more variability in how the female AI assistant was evaluated relative to the male AI assistant across different tasks. In the present work, we suggested such variability is a reflection of the degree to which women receive more societal punishment than men for not conforming to gender roles. It also might reflect double standards in that men are viewed as able to perform stereotypical feminine tasks, but the opposite is not true. These possibilities and others should be explored to better understand how gender roles and biases translate to human-AI interaction.

Finally, the study only manipulated pictures and names of the AI assistants and did not consider other factors such as voice that might trigger perceptions of gender (or the absence of gender cues). Future research could expand on this by also analyzing visual, vocal, and other characteristics to see if qualitatively different gender cues trigger similar reactions and role expectations. Moreover, such work should investigate whether participant gender influences these processes. Past findings are mixed regarding the extent to which a person's gender identity affects how they evaluate gendered AI. More work is needed to clarify such relationships.

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## Conclusion

The present study explored the interaction between the perceived gender of AI assistants and the stereotypical nature of their assigned tasks. Our findings suggest that gender-task alignment increases credibility perceptions for AI perceived to be feminine, but this effect is not observed for AI perceived to be masculine. Furthermore, the study offers important insight into how even superficial manipulations—such as the avatar and name of an AI assistant—can significantly influence perceptions of its credibility. Our findings add to a growing literature that seeks to explore how the perceived gender of AI influences user evaluations and the degree to which such effects are dependent on the type of task involved.

## CRedit Authorship Contribution Statement

**Joana Videira:** Conceptualization, Methodology, Formal Analysis, Investigation, Writing – Original Draft, Writing – Review & Editing, Project administration

**David C. DeAndrea** Conceptualization, Methodology, Formal Analysis, Investigation, Writing – Review & Editing, Supervision, Funding Acquisition

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